

RANDIKA PRABASHWARA

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PROFESSIONAL STATEMENT

Computer Science & Eng. graduate with hands-on Data Science & Engineering experience delivering production AI systems that drive measurable business impact. Expertise in Computer Vision, NLP, and Generative AI with strong foundation in MLOps, cloud deployment, and cross-functional collaboration. Passionate about transforming strategic visions into scalable, business-ready ML solutions.

EDUCATION

- B.Sc. Engineering (Honours)** Jun 2021 - Jun 2025
Department of Computer Science and Engineering, University of Moratuwa
 - Specialization in Data Science & Engineering
- Minor in Mathematics** Jun 2024 - Jun 2025
University of Moratuwa, Sri Lanka
 - Time Series and Stochastic Process, Linear Models and Multivariate Statistics
- G.C.E Advanced Level Examination (Physical Science Stream)** Jan 2010 - Aug 2019
Prince of Wales College, Moratuwa
 - Z-score: 2.1848 | Island Rank: 29 | All 'A' passes for Combined Mathematics, Physics, Chemistry

WORK EXPERIENCE

- Research Assistant** Jul 2025 - Present
University of Moratuwa
 - Advanced ML Research:** Developing hybrid deep learning architectures combining generative adversarial networks with physics-informed neural networks (PINNs) and diffusion-based probabilistic models for multi-scale super-resolution, incorporating domain-specific constraints through physics-aware loss formulations
 - Cross-disciplinary Collaboration:** Collaborating with faculty and graduate students on interdisciplinary projects spanning computer vision, multi-model learning, and financial AI
 - Technical Innovation:** Developing novel deep learning architectures and optimization algorithms, achieving significant improvements in model performance and computational efficiency across multiple research domains
- Data Science Engineer Intern** Dec 2023 - Jun 2024
Brown and Company PLC, Colombo 03
 - End-to-End ML Delivery:** Partnered with business stakeholders, product owners, and IT teams to translate ambiguous business requirements into production-ready ML solutions, delivering three major projects
 - Sales-Insight Navigator:** Led analytics development in cross-functional team to address field sales visibility gaps, building comprehensive platform with real-time GPS tracking, territory coverage analysis, performance scorecards, and opportunity identification, improving sales team efficiency by 35%
 - Cheque OCR System:** Collaborated within 3-member development team to eliminate manual data entry bottlenecks, contributing core computer vision algorithms using Google Cloud Vision API, Azure Computer Vision API, and custom vision models for mobile integration
 - Report Generation Automation:** Independently architected automated pipeline replacing manual Excel-based reporting, delivering data extraction, comprehensive EDA, and dynamic visualization generation to eliminate repetitive processes
- Teaching Assistant** Jan 2025 - Jun 2025
University of Moratuwa
 - Automated Assessment Development:** Developed and implemented automated evaluation system for RPAL programming assignments, streamlining assessment of 100+ student projects and reducing grading time by 70%

SKILLS

- ML Frameworks & Tools:** Python (4+ yrs), TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn, LangChain, Huggingface Transformers, LlamaIndex, pandas, NumPy, Jupyter
- MLOps & Deployment:** MLflow, Model Serving, CI/CD Pipelines, Model Monitoring A/B Testing, Docker, Model Optimization, Kubernetes, Feature Engineering at Scale
- API Development:** FastAPI, Flask, RESTful APIs, Model Deployment, Microservices Architecture
- Cloud & Infrastructure:** Google Cloud Platform (GCP), AWS (EC2, S3, Lambda), Azure, Apache Spark, PySpark, Hadoop, CUDA, Distributed Computing
- Data Engineering:** MySQL (3+ yrs), PostgreSQL, MongoDB, Pinecone(Vector DB), GraphDB, HBase, Redis, Data Pipeline Development, ETL/ELT Processes, Data Cleaning, Feature Engineering
- ML/AI Specializations:** Computer Vision, Natural Language Processing, Time-Series Forecasting, Generative AI, Multi-Modal Learning (Tabular, Text, Image), RAG Systems
- Software Engineering:** Java (3+ yrs), C++, JavaScript, ReactJS, Git, C#
- Core Competencies:** Deep Learning Architecture Design, Image Processing, Time-Series Analytics, Prototyping, Big Data Processing, Mathematical Modeling, Statistical Analysis

PROJECTS

Research Projects

- **CSM-SR: Conditional Structure-Informed Multi-Scale Super-Resolution GAN** Jun 2024 - Oct 2025
Final Year Research Project
 - **Problem & Gap:** Scientific imaging applications require ultra-high resolution reconstruction while preserving critical structural details. Traditional methods achieve only basic pixel enhancement, losing fine-grained features crucial for nanoscience and biomedical research.
 - **Solution:** Developed novel deep learning architecture integrating encoder-based feature conditioning, multi-scale discriminator design, and custom Semantic Structural Loss (SSL) function with three-stage pipeline.
 - **Achievements:**
 - 3.5dB PSNR improvement, 20% SSIM increase and 20% LPIPS decrease over state-of-the-art methods
 - 38.79 PSNR/0.978 SSIM/0.178 LPIPS on SEM datasets (2.8dB improvement in PSNR, 0.171 increase in SSIM, and 0.10 reduction in LPIPS over competitors)
 - Outperformed EDSR, ESRGAN, SwinIR, HMANet on benchmark datasets (Set5, Set14, BSD100)
 - **Tech Stack:** *Python, TensorFlow, Keras, PyTorch, CUDA, OpenCV*
 - **Links:** [[GitHub Repository](#)] — [[Interactive Demo](#)] — [[Supplementary Materials](#)] — [[Dataset & Benchmarks](#)]
- **MedAssist AI: Multi-Model RAG-Powered Medical Conversational System** Aug 2025 - Present
 - **Problem & Gap:** Medical information systems lack intelligent query processing and source attribution capabilities. Traditional systems provide generic responses without leveraging comprehensive knowledge bases or maintaining conversation context.
 - **Solution:** Built RAG-powered medical chatbot with multi-model LLM integration (Llama, OpenAI, Gemini, Claude), vector database for semantic search, and intelligent query processing with conversation memory.
 - **Achievements:**
 - Multi-model consensus routing with 80% query classification accuracy
 - **Tech Stack:** *Python, FastAPI, LangChain, PyTorch, Pinecone, OpenAI API, Docker, CUDA*
 - **Links:** [[GitHub Repository](#)]

Enterprise Projects

- **GeoSales Intelligence Platform** Dec 2024 - Oct 2025
 - **Business Context:** Partnered with sales stakeholders to translate vague requirement ("better field visibility") into concrete ML solution: GPS-based route analysis, territory mapping, and predictive opportunity scoring.
 - **End-to-End Delivery:** Built complete ML pipeline from scoping and data processing through deployment and monitoring, processing real-time GPS streams with geospatial-temporal clustering and predictive analytics
 - **Technical Implementation:**
 - Processed GPS trace data to reconstruct Business Process Officer (BPO) movement patterns and routes
 - Implemented geospatial-time series clustering algorithms (DBSCAN, Optics) to identify natural sales territories and customer clusters
 - Built feature engineering pipeline extracting temporal patterns and territory coverage metrics from GPS data
 - Developed FastAPI microservices for real-time GPS stream processing with < 500ms latency
 - **Business Impact:**
 - Achieved < 500ms latency with 88% prediction accuracy in production
 - 35% increase in sales efficiency, 40% reduction in operational costs
 - Proactively identified and resolved data quality issues, preventing 2-week delay
 - **Tech Stack:** *Python, FastAPI, Apache Kafka, PostgreSQL, Redis, React, Docker, Azure, Git*
 - **Links:** [[GitHub Repository](#)]
- **Advanced Sri Lankan Driving License OCR System** Apr 2025 - May 2025
 - **Challenge:** Existing OCR solutions struggle with Sri Lankan driving licenses: complex table structures, varying orientations, poor image quality, achieving only 60-70% accuracy.
 - **Solution:** Revolutionary multi-model OCR pipeline combining TrOCR, EasyOCR, and Enhanced Tesseract with intelligent multi-factor orientation detection and adaptive image enhancement.
 - **Achievements:**
 - 96% accuracy in vehicle class extraction, 92% accuracy in date extraction
 - 1.8 seconds average processing time per license
 - 4-factor orientation detection with 98% success rate
 - **Tech Stack:** *TrOCR, EasyOCR, PyTorch, OpenCV, Transformers, CUDA*
 - **Links:** [[GitHub Repository](#)]

Big Data Projects

- **Music Genre Classification Engine** Mar 2025 - Jun 2025
 - Built Spark-native distributed machine learning pipeline with ensemble approach combining Random Forest, Gradient Boosting Trees, and Logistic Regression
 - Implemented scalable feature extraction and model training across distributed clusters
 - **Tech Stack:** *Python, Apache Spark, PySpark MLlib, Audio Processing, MLOps*
 - **Links:** [[GitHub Repository](#)] — [[Live Demo](#)]

PUBLICATIONS

A Data-Driven Spatiotemporal Framework for Retail Analytics. In Applied Data Science & Artificial Intelligence (AD-ScAI) Summit 2025

CERTIFICATIONS

- **AWS:** AWS Academy Graduate - [Data Engineering](#), [Machine Learning for NLP](#), [ML Foundations](#), [ML Specialization](#)
- **NVIDIA:** NVIDIA Deep Learning Institute - [Fundamentals of Deep Learning](#)
- **Stanford:** Stanford University (Coursera) - [Machine Learning Specialization](#)
- **DeepLearning.AI:** Mathematics for Machine Learning & Data Science - [DeepLearning.AI](#)

HIGHLIGHTS

- AMP@-Parkinson's Disease Progression Prediction - Kaggle: Top 83%
- Sri Lankan Mathematical Olympiad: High distinction pass - 2011/12/13/14/15/17
- Award for the Best Result in the Physical Science Stream - Prince of Wales College, Moratuwa

REFERENCES

Dr. Uthayasanker Thayasivam

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